Kerbo-S Sprint 2 Reflection Report

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Accomplishments

Sprint 2 ran one week, from Sun Nov 1 to Sat Nov 7. Its goal was to turn the Sprint 1 design into a working, tested audit engine running on real MIT Kerberos data. Every Sprint 2 code card is complete and merged, and the full test suite passes.

A real Kerberos lab and real test data

- **Lab KDC built** (Ben, Thu Nov 5): MIT Kerberos on Ubuntu Server 26.04 LTS as a Proxmox VM, realm KERBO.LAB. TC-ENV-01 passed: `kinit alice` and `klist` show a ticket-granting ticket. The setup is recorded in `docs/lab-setup.md` (PR #10).
- **Three fixture sets captured from the lab** with a capture script (Ben, PRs #12 and #14): a clean realm, a planted realm with eight deliberate weaknesses, and an aged copy for the password-age and inactivity rules. Each has an `expected_findings.yaml` written before the audit ran, including boundary cases that must not be reported.

Working code (pull requests #6 to #18)

- **Data models and parsers** for `getprinc`, `getpol`, `krb5.conf` and `kdc.conf`, with a loader that reads a whole fixture folder (David, PRs #6 and #7; Jessica, PR #9; Ben, PR #15).
- **Eight audit rules**: KC-01 to KC-04 (Jessica, PR #13; KC-01 from Sprint 1) and KC-06 to KC-09 (Jessica, PR #9), each with boundary tests such as 91 days flagged and 90 not.
- **Risk scoring** for all eight rules, matching the Requirements Section 6.2 values exactly (KC-07 88.5 Critical, KC-02 70.0 High).
- **SQLite storage** with five tables and database triggers that reject any change to a saved finding (Ben, PR #16).
- **A findings window** with severity and category filters, counts by severity and clear error messages, and a **command-line audit** that prints JSON and saves every run (Jessica, PR #13; Ben, PR #16).

- **End-to-end tests on the lab data** (Ben, PR #17): clean realm 0 findings on two identical runs, planted realm 8 of 8 expected findings, aged realm 11 of 11, with no extra findings and nothing from the not-expected lists.

Testing and evidence

- **68 automated tests pass** on the final main branch (`docs/evidence/pytest-sprint2.txt`, PR #18), up from 14 at the end of Sprint 1.
- **24 of 59 test plan cases Pass, none Fail.** All 35 not yet tested are Sprint 3 work. Screenshots for the manual cases are in `docs/evidence/`.
- **Two real defects found and fixed** before the final run, both only visible on real lab output: RC4 keys labelled `DEPRECATED:arcfour-hmac` were being missed by KC-07, and an `includedir` line in the lab `krb5.conf` was being reported as a parse error (fixed in PR #13).

Documents

The five Sprint 2 documents were updated from their Sprint 1 versions: Requirements v2.1 now has a status for every requirement (12 Complete, 20 Partial, 22 Outstanding, the Outstanding ones mainly the AI and reporting work moved to Sprint 3); Design v2.1 describes the code as built with 13 new design decisions; the Change Document records 22 new Sprint 2 changes; and the Test Plan shows real results.

Progress Since Sprint 1

Sprint 1 ended with a design and a 14-test skeleton running one rule on a hand-written sample. Sprint 2 ends with eight rules running on real KDC output, scored, stored and displayed.

Sprint 1 end (Nov 3)	Sprint 2 end (Nov 7)
No lab; TC-ENV-01 not run	Lab KDC on Proxmox; TC-ENV-01 Pass
One rule (KC-01)	Eight rules (KC-01 to KC-04, KC-06 to KC-09)
Hand-written sample snapshot	Clean, planted and aged fixtures captured from the lab, with manifests
No parsers	Parsers for principals, policies and both configuration files
No database	SQLite with immutable findings

Sprint 1 end (Nov 3)	Sprint 2 end (Nov 7)
14 tests	68 tests
0 of 57 test cases run	24 of 59 Pass, 0 Fail
5 pull requests	18 pull requests

Were the Sprint 2 goals met? The Sprint 1 reflection set four checks for success:

Goal from the Sprint 1 reflection	Result
pytest passes on every teammate's machine	Partly. The suite passes on Ben's Linux development VM (final log) and Jessica's Windows laptop. Chance and David ran the 14-test starter in Sprint 2 but did not record a run of the final suite.
The audit runs against real lab output	Met: clean, planted and aged realms (TC-KC-12, TC-KC-15)
Findings are saved to SQLite and cannot be edited	Met: TC-DB-01 and TC-DB-02 Pass
The test plan shows real Pass or Fail results	Met: 24 Pass with evidence

The scope set at the start of Sprint 2 (lab, real data, rules KC-01 to KC-09 except KC-05, immutable storage, findings table with filters and error messages, evidence-backed results) was delivered in full. KC-05 moved to Sprint 3 because it needs the KDC log. The demonstration did not happen: it is on hold at the instructor's direction.

Sprint 1 recovery items. All closed: the Sprint 1 package was resubmitted and accepted (Nov 3); the repository is public; the old branch and `.gitignore` were cleaned up (R-02, PR #8) and the Sprint 1 PDFs added to the repository (R-07, PRs #4 and #5), both done by Jessica; TC-ENV-01 passed (Nov 5).

Use of AI

We used Claude (Anthropic) as a drafting, coding and checking assistant. Kerbo-S itself still uses no AI: the AI Gateway is Sprint 3 work. This section records the use we know of, which was mainly through the coordinator's account.

Use	What Claude did	How we checked it
Planning	Drafted the one-week Sprint 2 plan and a step-by-step guide for each member	Each guide was matched to the Trello cards and their "Done when" lists
Code	Drafted the configuration parser and KC-06 to KC-09 (PR #9), KC-02 to KC-04, the snapshot errors and the findings table (PR #13)	Run on a real machine against the full test suite, and against Ben's real lab fixtures before merging; the window was checked by hand on Jessica's laptop
Pull request review	Pulled each pull request, ran the tests and checked it merged cleanly, for example Ben's PRs #12 and #14 to #18	The test counts it reported (28, 53, 68) were confirmed by the evidence log
Daily record	Checked the repository, Trello and the shared documents and wrote factual entries	Times and pull request numbers were checked against GitHub and Trello
Documents	Drafted the five Sprint 2 documents from the record, the merged code and the test results	Read against the rubric, the repository and the test log

Where we did not simply trust the AI.

- **Real data beat the AI's assumptions.** The AI-drafted KC-07 passed its own tests but missed the planted RC4 key, because the lab's MIT version labels it `DEPRECATED:arcfour-hmac`. The AI-drafted configuration parser also rejected the `includedir` line in the real `krb5.conf`. Both were caught only because the code ran against Ben's captured lab output, which is why we test against real fixtures and not only against AI-written samples.
- **The AI got a repository fact wrong.** On Nov 5 Claude reported that Ben's PR #10 did not exist. A screenshot of GitHub showed it did; the AI's check had run before GitHub's list updated. We now confirm repository facts on GitHub itself before acting on them.
- **The window could not be tested by the AI.** PyQt6 could not be installed in the AI's workspace, so it tested the window logic against a stand-in. That found one real bug (a filter callback with the wrong signature), but the actual window was only trusted after a person opened it and took the TC-UI-02 and TC-UI-06 screenshots.

What Worked Well

- **Testing on real lab output.** The two defects above would have reached Sprint 3 with hand-written samples. Capturing the real `kadmin.local` output and committing it also let every member run the full suite without access to the lab.
- **Writing the expected findings first.** Because the manifests existed before the audit ran, "8 of 8" and "11 of 11" are real measurements, and the boundary cases (a password exactly 90 days old, a lifetime of exactly 24 hours) prove the rules do not over-report.
- **Small pull requests with the test count in each one.** Every code pull request said how many tests passed, so a reviewer could check it in minutes. 18 pull requests merged in Sprints 1 and 2.
- **A dated daily record.** Keeping one line per person per day, with times and pull request numbers, gave an objective basis for the instructor meeting and for this report, instead of impressions.
- **Pulling together at the end.** When the code freeze passed with cards unfinished, the team redistributed the remaining work on Friday night and finished every Sprint 2 code card by 1:31 AM Saturday instead of carrying a backlog into Sprint 3.

What Needs Improvement

- **Work started late and landed at the end.** The lab, which every data-dependent card needed, was not started until Thu Nov 5, the day of the planned code freeze. Real fixtures arrived Friday night, so the end-to-end work ran into Saturday morning. Ben's own note proposes the fix: finish anything others depend on by Monday of each sprint.
- **Uneven participation.** From Nov 1 to Nov 4 Ben had no activity, and Chance's only commit was his starter test run (PR #2). Chance's coding cards and part of David's were completed by others under the Friday redistribution (see the record below). Participation was the subject of a mandatory team meeting with Dr. Zarafshani on Thu Nov 5.
- **Pull request discipline.** Five commits went straight to the main branch without a pull request (evidence and screenshot files only), and one pull request was merged by its author before review. Every change, including screenshots, should go through a pull request.
- **A credential was posted in a shared document.** A lab VM password was posted in a document comment on Nov 5. The team rule since Nov 6 is that credentials go only through Microsoft Teams. The comment can only be deleted by its author, and the

password should be changed.

- **Nobody commented on the documents.** R-05 (every member comments on the Sprint 1 documents) closed with no comments, and the Sprint 2 documents were drafted by one member. Reviewing each other's documents should be a card with an owner and a due date.
- **No automatic test run.** Tests run only when someone runs them. A GitHub Actions workflow would test every pull request.
- **The demonstration did not happen.** It is on hold at the instructor's direction.

Team participation record (Sprint 2, Nov 1 to 7, 2026)

Taken from the daily record, the GitHub repository and the Trello board. All times Central.

Person	What happened	Missing or done by others
Ben	Nov 1 to 4: no commits, pull requests or Trello activity. Thu Nov 5: built the lab KDC (S2-00), ran TC-ENV-01 (Pass), pushed two evidence commits straight to main (1:08 and 1:10 PM), opened PR #10 (lab setup note and admin fixture, merged by Jessica 5:25 PM). Fri Nov 6: sent his Sprint 2 notes (about 9:40 PM); pushed the clean fixtures with a capture script (PR #12, 9:54 to 10:04 PM); reported the two defects found on real lab data; planted and aged fixtures (PR #14, merged 11:40 PM). After the 10:40 PM redistribution, took David's remaining parts. Sat Nov 7: loader finish (PR #15), SQLite storage (PR #16, merged by himself), end-to-end tests (PR #17) and the test evidence (PR #18, 1:22 AM); one screenshot pushed straight to main (12:42 AM).	His S2-01 starter pull request was never opened. R-04 (TC-ENV-01) and S2-00 finished two days late. The S2-23 test results due Fri 12 PM arrived Sat 1:22 AM. Posted the lab VM password in a document comment (Thu 12:55 PM).

Person	What happened	Missing or done by others
Chance	Mon Nov 2: ran the starter and opened PR #2 (S2-01, 14 tests pass). Thu Nov 5: a test comment on the Requirements document; marked R-05 and S2-01 complete on Trello; asked for access to three documents. Fri Nov 6: sent his Sprint 2 notes in the group chat (about 3:45 PM).	No code for his cards S2-08, S2-14 and S2-16, or for R-02 and R-07. All five were completed by Jessica (PRs #4, #5, #8, #13). No activity Sat Nov 7.
David	Mon Nov 2: ran the starter, PR #3 (14 tests pass). Wed Nov 4: data models, PR #6 (5:00 PM); principal and policy parsers with the fixture loader and four tests (branch pushed 6:01 PM; PR #7 opened and merged by Jessica Thu 4:28 AM). Thu Nov 5: attended the team meeting; reviewed and merged Jessica's PR #9 (12:50 PM); posted his Sprint 2 notes (6:04 PM). Fri Nov 6: asked about the fixtures he was waiting on (about 2:10 PM).	His remaining parts (the loader finish in S2-06, S2-12 and S2-22) were completed by Ben after the redistribution.
Jessica	Coordinated the sprint: wrote the Sprint 2 plan and each member's guide (Nov 1), kept the daily record, and reviewed and merged pull requests #2, #3, #7, #10, #12, #14, #15, #17 and #18. Code: the S2-09 configuration parser and KC-06 to KC-09 (PR #9); R-07 and R-02 for Chance (PRs #4, #5, #8); the frozen-code test evidence (PR #11); Chance's S2-08, S2-14 and S2-16 plus the two defect fixes (PR #13, Fri 11:21 PM). Wrote the meeting brief for Thu Nov 5 and the Sprint 2 documents. Two screenshot renames pushed straight to main by accident (Fri 11:30 to 11:38 PM).	None for Sprint 2.

Team meeting, Thu Nov 5, 12:15 PM. A mandatory meeting with Dr. Zarafshani about participation. Outcome: the team stays the same for now, and Chance and Ben are given another opportunity to change how they have been participating. Grades are individual.

Redistribution, Fri Nov 6, about 10:40 PM. To finish Sprint 2 completely by the Saturday submission, the remaining work was redistributed: Ben finished the planted and aged fixtures and took David's remaining parts, and Jessica took Chance's remaining parts. This was because of time constraints, not because David and Chance did not want to do the work or did not do their part. The team pulled together to fully complete Sprint 2. On Trello each card keeps its original owner and names who did it, for example "[Owner: David; done by Ben]".

Members' own reflections (from their Sprint 2 notes). Ben: "In Sprint 3 I'll start the lab work earlier so nobody is waiting on the lab or fixtures." Chance: "In the next sprint I shall begin the implementation earlier so that any issues I face, I will have more time to solve and troubleshoot them." David noted that validation on real KDC fixtures was pending until the files were available; it was completed with Ben's fixtures in PRs #15 and #17.

Focus for Sprint 3

Sprint 3 completes the MVP. No Sprint 2 code carries over, so Sprint 3 starts directly on the work moved there on purpose.

Area	Work	Test cases
Logs and ACL	KDC log and `kadm5.acl` parsers (S2-07); rules KC-05, KC-10, KC-11; KC-06 for ACL entries	TC-COL-04, 05; TC-KC-05, 10, 11
Collection	Live collector with the command allow-list; snapshot export; hashing at capture and checking on import	TC-COL-06, 08, 09; TC-ENV-03
AI	AI Gateway with the mock and OpenAI providers; choose the model; the Section 7.2 evaluation	TC-AI-01 to 08; TC-EVAL-02
Reports	Technical and executive reports, PDF export, digest and verify	TC-RPT-01 to 04

Area	Work	Test cases
Interface	Audit progress on a worker thread, finding detail and review, live-audit errors	TC-UI-01, 03, 04
Quality	Compound scenario TV-15, the metric evaluation script, security and non-functional tests	TC-KC-14; TC-EVAL-01; TC-SEC, PERF, USE, PORT, MNT
Demo	Demonstration once the instructor sets a date	

What we will change in how we work.

1. **Dependencies first.** Anything another member needs (lab data, fixtures, shared models) is due on the Monday of the sprint, as Ben proposed.
2. **Every change through a pull request,** including screenshots and evidence, reviewed by someone other than the author before it is merged.
3. **Automatic tests on every pull request** with a GitHub Actions workflow.
4. **Document review as a card.** Each member reviews one Sprint 3 document and leaves comments by a set date.
5. **Credentials only through Microsoft Teams.**
6. **Keep the daily record,** so the Sprint 3 reflection is built from facts.

How we will know it worked. By the end of Sprint 3, all eleven rules run on real lab data including the logs, the AI explanations work with a tested fallback, a PDF report exports and verifies, the metric script reports detection and false-positive rates against the manifests, and every member's work appears in the repository through their own pull requests.